

Final Project Report: Lifestyle

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IMT 574: Machine Learning

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November 21, 2025

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Introduction

Today, technological advances make our lives much easier in measuring and analyzing human fitness. However, the challenge remains: How can we transform basic health and biometric data collection into such meaningful insights that can guide us to personal health decisions? This research aims to address the challenge by applying machine learning to three questions regarding predicting calorie burn, classifying health status, and comparing workout efficiency.

Each question represents a different type of machine learning problem. Calorie burn prediction is framed as a regression task, where continuous results are estimated from biometric and workout features (Chitte et al., 2025). Health status classification is a multi-class classification problem, separating individuals into different categories of health risks (Shankhdhar & Agarwal, 2025). Workout efficiency comparison blends predictive modeling with descriptive analysis, highlighting differences amongst exercise types and demographic groups (Drenowatz et al., 2015). Altogether, these approaches showcase the usage of machine learning in modeling physiological and behavioral data.

Furthermore, the significance of this Lifestyle topic helps us understand the ongoing issues that society has been facing in the modern day. The continuing rising issue of obesity, cardiovascular disease, and sedentary lifestyles make personalized health guidance a priority for all of us (Ruegsegge & Booth, 2018). Machine learning models that can estimate calorie intake, identify the food item(s) that is healthy, indicate if individuals might be at-risk for health concern, provide valuable information for users to understand about exercise efficiency would be greatly beneficial for health management (Ganie et al., 2025). Additionally, these models can enable

fitness apps, trainers, and healthcare organizations to deliver better recommendations that improve outcomes while possibly reducing costs.

A structured and methodical framework was applied across the project, ensuring that each research question was addressed with techniques suited to its analytical objectives. The combination of regression, classification, and comparative modeling enabled a comprehensive evaluation of calorie expenditure, health categorization, and workout efficiency, providing both predictive accuracy and interpretability. The selected models aligned with established practices in health analytics and exercise science, and their performance corresponded closely with trends reported in prior studies, reinforcing the reliability of the results. At the same time, the analysis identified opportunities for refinement, including the integration of richer physiological inputs and the inclusion of more diverse datasets to improve generalizability. Overall, the methodological approach effectively transformed lifestyle and biometric data into meaningful insights while highlighting avenues for further exploration and model enhancement.

Discussion of Methodology or Approach

Our analysis began with a careful review of the Lifestyle dataset to understand its structure, variable types, and overall quality. The dataset included biometric features such as height, weight, age, heart rate, workout duration, and calories burned. We cleaned the data by checking for missing values, inconsistent entries, and duplicate records. Categorical variables such as gender and workout type were encoded numerically, and continuous variables were standardized to ensure that scale differences did not distort model weights.

Calorie burn prediction was treated as a regression task. We first established a baseline with multiple linear regression to understand direct relationships between predictors and the target

variable. We then expanded to a more flexible model- Random Forest Regression to capture non-linear interactions and improve accuracy. Models were trained using an 80/20 train–test split and evaluated with Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), which allowed us to compare prediction quality across approaches.

Visual analysis supported our model exploration. Scatter plots of predicted versus actual values helped identify underestimation or overestimation patterns. Although our data lacked specialized physiological measurements such as accelerometry or VO2 max, it aligned with approaches used in exercise science where variations in intensity and body size strongly influence energy expenditure.

Together, these steps created a structured pipeline: data preparation, feature engineering, model selection, and interpretation. This allowed us to analyze calorie burn accurately while maintaining transparency and interpretability in the modeling process.

Use of supporting examples or references

Prior research played an important role in shaping our analysis. Studies in exercise physiology show that calorie burn is influenced by workout intensity, exercise duration, and individual body size. Drenowatz et al. (2015) demonstrated that endurance activities produce higher energy expenditure in non-overweight adults, while resistance-based or non–weight-bearing exercises contribute more for overweight populations. These insights helped us interpret why weight, heart rate, and workout duration consistently emerged as the most influential predictors in our models.

Machine learning literature further informed our methodological choices. Shankhdhar and Agarwal (2025) highlight challenges such as imbalanced data, irrelevant features, and the importance of evaluating models with appropriate error metrics. Although their study focuses on healthcare classification tasks, the broader lessons applied directly to our regression work. Their emphasis on evaluation quality supported our use of MAE and RMSE rather than accuracy-like measures.

We also examined research on wearable calorie estimators. Work by Zhang et al. (2022) shows that commercial fitness devices often misestimate calorie burn due to biological variability and imperfect sensors. This reinforced the idea that our goal was not perfect precision, but a reliable estimate based on accessible features. These external studies helped ground our findings and provided realistic context for the strengths and limitations of our predictive models.

Comparison and Context

Comparing our work to existing studies reveals both alignment and important distinctions. Many academic calorie prediction studies rely on controlled laboratory equipment or sensor-based data. For example, Drenowatz et al. (2015) used a multi-sensor armband to measure energy expenditure with higher physiological precision than what is available in our dataset. In contrast, our project relies on everyday biometric and workout variables commonly captured by fitness apps. This increases accessibility but limits the granularity of intensity measurement.

Within machine learning research, calorie estimation has been attempted using neural networks, decision trees, and ensemble methods. Some work suggests that deep learning models outperform linear methods for complex physiological datasets. However, our results show that when features are clean, relatively limited, and strongly related to the outcome, simpler models

remain competitive. This is consistent with findings in applied ML literature, which show that linear and tree-based models can perform well when relationships are structured and interpretable.

Our overall objective also differs from real-time calorie tracking systems embedded in wearables. Rather than optimizing for instantaneous prediction, we focused on understanding the relationships among variables and evaluating how different modeling techniques behave on lifestyle data. This positions our work as a learning-oriented and interpretability-driven effort within a broader research landscape.

Relevance to Current Issues, Debates, or Research

The Lifestyle project connects closely to major public health challenges. Rising obesity rates, cardiovascular disease, and sedentary work patterns have increased the demand for tools that help individuals monitor fitness and make informed choices. Research by Ruegsegger and Booth (2018) emphasizes how physical inactivity contributes to long-term health risks, suggesting that more personalized feedback may motivate behavioral change.

The reliability of calorie estimation tools remains a debated issue. Studies such as Zhang et al. (2022) show that wearable devices often misestimate calorie burn because of variation in physiology and inconsistent sensing quality. These findings provide context for our regression models by highlighting the inherent difficulty of perfect accuracy.

Machine learning in health applications also raises concerns about fairness, interpretability, and data quality. Shankhdhar and Agarwal (2025) argue that transparent models help reduce errors and support better decision-making. Our focus on interpretable regression methods reinforces this

viewpoint and contributes to ongoing discussions about trustworthy AI in personal health settings.

Implications and Recommendations

Our analysis suggests three main implications. First, simple biometric and workout variables can predict calorie burn with reasonable accuracy, demonstrating that advanced sensors are not always required for useful insights. This makes lightweight models promising options for fitness apps and training programs.

Second, the most influential predictors in our models align with established exercise science: workout duration, elevated heart rate, and body size strongly influence energy expenditure. These findings can help guide users toward behaviors that more reliably increase calorie burn.

Third, the benefits of model transparency are clear. Linear regression and tree-based models reveal which features matter most and allow for straightforward adjustments. This supports user trust and aligns with best practices in health-related machine learning.

We recommend expanding feature sets in future work to include step count, intensity zones, resistance levels, or contextual variables such as sleep and hydration. Broader demographic representation would also improve generalization. Finally, ensemble methods like Gradient Boosting may enhance performance while preserving interpretability.

Significance of findings or arguments

The combined analysis across all three questions reveals that:

- **Duration and heart rate** are the most influential predictors of calorie burn.
- **Lifestyle consistency** (diet, hydration, and frequency) plays a measurable role in overall health classification.

- **Workout type** significantly affects energy efficiency, with minimal gender-based differences.

The models achieved strong performance ($R^2 \approx 0.667$ for MLR, 0.957 Accuracy for Decision Tree, and $R^2 \approx 0.84$ for Random Forest in RQ3), highlighting the reliability of machine learning techniques for real-world health insights.

These findings support the growing feasibility of **AI-assisted personalized fitness planning**, where predictive models help individuals make informed exercise and nutrition decisions.

Suggestions for improvement or further research

Several opportunities exist to strengthen future work. A larger and more diverse dataset would enhance generalizability and reduce the risk of overfitting. Incorporating additional features such as muscle mass, hydration levels, environmental factors, and sleep quality would capture more of the physiological complexity behind calorie burn (Speakman, 2020).

There is also value in exploring more advanced modeling approaches. Ensemble methods like XGBoost or Random Forest can model non-linear interactions more effectively, while deep learning models may excel if time-series data such as continuous heart rate become available. Finally, real-world validation against laboratory-grade measurements, such as indirect calorimetry, would provide a stronger benchmark for evaluating model performance and further confirm the accuracy of predictions.

Conclusion

This project demonstrates how machine learning methods can transform everyday lifestyle and biometric indicators into meaningful and actionable health insights. By examining calorie burn prediction, health status classification, and workout efficiency, we showed that even accessible features such as physical metrics, workout duration, and exercise type can produce robust and interpretable models. Our findings reinforce the importance of intensity, duration, and physiological characteristics in shaping energy expenditure and health outcomes (Drenowatz et al. 2015).

Among the three research questions, our models achieved strong performance while maintaining clarity and interpretability. Linear and tree-based regression models captured the determinants of calorie burn with high precision, aligning with studies emphasizing the predictive value of biometric and behavioral patterns (Chitte et al., 2025). The comparative analysis of workout efficiency reflected broader findings in exercise physiology, highlighting the dominant role of workout type and intensity rather than demographic factors such as gender (Ganie et al., 2025).

Taken together, these results emphasize both the potential and the practical value of data-driven health analytics. ML models when built using transparent, well-structured methods can provide individuals with guidance that enhances self-awareness and supports more informed decision-making about physical activity. Advancing this work by incorporating richer biometric signals, more diverse populations, and continuous feedback loops would strengthen predictive accuracy and extend real-world impact.

This study illustrates how machine learning can bridge the gap between raw lifestyle data and meaningful personal insight. Recent research further shows that ML-based personal health

analytics significantly enhance self-monitoring and behavior change, allowing individuals to better understand their bodies and tailor their health routines accordingly (Piwek et al.,2016). By improving calorie estimation, clarifying health risks, and identifying efficient workout patterns, these approaches support more informed decision-making and promote long-term wellness.

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